POLITECHNIKA WROCŁAWSKA Wydział Informatyki i Telekomunikacji	Wydział: Informatyki i Telekomunikacji Kierunek: Cyberbezpieczeństwo Rok Akademicki: 2025/2026 Rok studiów, semestr: *2, 2* Grupa: *2*
Ochrona Systemów Operacyjnych – Laboratorium 5	
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Zadanie 1

```
root@Ubuntu:/home/wesoly# systemctl start ssh
root@Ubuntu:/home/wesoly# systemctl start ntp
root@Ubuntu:/home/wesoly# systemctl start postgresql
root@Ubuntu:/home/wesoly# systemctl start apache2
```

Rysunek 1 Włączenie usług

1. Proszę wyświetlić listę działających usług przy pomocy „ss”.

```
root@Ubuntu:/home/wesoly# ss -ntlup
Netid  State   Recv-Q  Send-Q                    Local Address:Port    Peer Address:Port  Process
udp    UNCONN  0       0                               0.0.0.0:36890        0.0.0.0:*      users:(("avahi-daemon",pid=961,fd=14))
udp    UNCONN  0       0                            127.0.0.54:53           0.0.0.0:*      users:(("systemd-resolve",pid=590,fd=16))
udp    UNCONN  0       0                         127.0.0.53%lo:53           0.0.0.0:*      users:(("systemd-resolve",pid=590,fd=14))
udp    UNCONN  0       0                       192.168.222.137:123          0.0.0.0:*      users:(("ntpd",pid=10874,fd=19))
udp    UNCONN  0       0                           127.0.0.1:123            0.0.0.0:*      users:(("ntpd",pid=10874,fd=18))
udp    UNCONN  0       0                             0.0.0.0:123            0.0.0.0:*      users:(("ntpd",pid=10874,fd=17))
udp    UNCONN  0       0                             0.0.0.0:5353           0.0.0.0:*      users:(("avahi-daemon",pid=961,fd=12))
udp    UNCONN  0       0        [fe80::20c:29ff:fea2:32e0]%ens33:123             [::]:*      users:(("ntpd",pid=10874,fd=21))
udp    UNCONN  0       0                                [::1]:123             [::]:*      users:(("ntpd",pid=10874,fd=20))
udp    UNCONN  0       0                                 [::]:123             [::]:*      users:(("ntpd",pid=10874,fd=16))
udp    UNCONN  0       0                                 [::]:5353            [::]:*      users:(("avahi-daemon",pid=961,fd=13))
udp    UNCONN  0       0                                 [::]:60704           [::]:*      users:(("avahi-daemon",pid=961,fd=15))
tcp    LISTEN  0       70                          127.0.0.1:33060        0.0.0.0:*      users:(("mysqld",pid=15215,fd=21))
tcp    LISTEN  0       4096                        127.0.0.1:631          0.0.0.0:*      users:(("cupsd",pid=7016,fd=7))
tcp    LISTEN  0       151                         127.0.0.1:3306         0.0.0.0:*      users:(("mysqld",pid=15215,fd=23))
tcp    LISTEN  0       4096                          0.0.0.0:22           0.0.0.0:*      users:(("sshd",pid=14972,fd=3),("systemd",pid=1,fd=155))
tcp    LISTEN  0       4096                      127.0.0.53%lo:53           0.0.0.0:*      users:(("systemd-resolve",pid=590,fd=15))
tcp    LISTEN  0       200                         127.0.0.1:5432         0.0.0.0:*      users:(("postgres",pid=13501,fd=6))
tcp    LISTEN  0       4096                         127.0.0.54:53           0.0.0.0:*      users:(("systemd-resolve",pid=590,fd=17))
tcp    LISTEN  0       4096                             [::1]:631            [::]:*      users:(("cupsd",pid=7016,fd=6))
tcp    LISTEN  0       511                                  *:80                 *:*      users:(("apache2",pid=14398,fd=4),("apache2",pid=14397,fd=4),("apache2",pid=14395,fd=4))
tcp    LISTEN  0       4096                              [::]:22              [::]:*      users:(("sshd",pid=14972,fd=4),("systemd",pid=1,fd=159))
```

Rysunek 2 Lista działających usług

Ataki są możliwe za pomocą ssh i apache2.

2. Proszę sprawdzić listę otwartych portów przy pomocy programu „nmap” (zainstalować jeśli trzeba) i porównać z listą otrzymaną poleceniem „ss”

```
root@Ubuntu:/home/wesoly# nmap localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 16:33 CEST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000070s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
631/tcp  open  ipp
3306/tcp open  mysql
5432/tcp open  postgresql
```

Rysunek 3 Wynik nmap’a

Wyniki są takie same.

3. Należy sprawdzić ustawienia firewalla oraz określić, które usługi powinny mieć ograniczony dostęp ze względu na bezpieczeństwo (proszę przyjąć jakieś hipotetyczne założenia).

Założenie: Nasz ubuntu jest serwerem, który świadczy usługi(stronę internetową) do Internetu.

```
root@Ubuntu:/home/wesoly# ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip
```

Rysunek 4 Status firewalla

Nasz firewall jest włączony, ale nie ma skonfigurowanych żadnych reguł oprócz domyślnych:

- Blokowanie ruchu przychodzącego
- Zezwalanie na ruch wychodzący
- Blokowanie routowania

W kontekście naszego serwera, powinniśmy dodać reguły:

- Zezwolenia na ruch przychodzący na port 80
- Zezwolenie na ssh tylko z konkretnego IP
- Pomimo, że postgresql,mysql,cupsd nasłuchują jedynie lokalnie, to je również możemy ograniczyć.

4. Proszę sprawdzić jakie są wymagania dotyczące haseł w systemie i zaproponować ich poprawę.

```
password        requisite                       pam_pwquality.so retry=3 minlen=12
password        [success=2 default=ignore]      pam_unix.so obscure use_authtok try_first_pass yescrypt remember=5 sha512 rounds=10000
password        sufficient                      pam_sss.so use_authtok
# here's the fallback if no module succeeds
password        requisite                       pam_deny.so
```

Rysunek 5 Zawartość pliku common-password

Ubuntu korzysta z modułu PAM, dlatego wszelkie ograniczenia dot. Haseł możemy ustawić w pliku /etc/pam.d/common-password
Nasz plik zawiera jedynie wpis *retry=3*, więc użytkownik ma 3 próby wpisania hasła.
Aby poprawić zabezpieczenia możemy do pliku dopisać kilka innych wymogów np. :

- Minlen=14
- Ucredit=-1
- Lcredit=-1
- Dcredit=-1
- Ocredit=-1
- Remember=5

```
root@Ubuntu:/etc/pam.d# systemctl status unattended-upgrades
● unattended-upgrades.service - Unattended Upgrades Shutdown
     Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; pre>
     Active: active (running) since Tue 2026-04-07 12:15:39 CEST; 1 week 2 days ago
       Docs: man:unattended-upgrade(8)
   Main PID: 1469 (unattended-upgr)
      Tasks: 2 (limit: 3921)
     Memory: 10.2M (peak: 11.4M)
        CPU: 111ms
     CGroup: /system.slice/unattended-upgrades.service
             └─1469 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade>

Apr 07 12:15:39 Ubuntu systemd[1]: Started unattended-upgrades.service - Unattended Up>
root@Ubuntu:/etc/pam.d# cat /etc/apt/apt.conf.d/20a
20apt-esm-hook.conf  20archive            20auto-upgrades
root@Ubuntu:/etc/pam.d# cat /etc/apt/apt.conf.d/20auto-upgrades
APT::Periodic::Update-Package-Lists "1";
APT::Periodic::Unattended-Upgrade "1";
root@Ubuntu:/etc/pam.d#
```

Rysunek 6

W systemie są uruchomione automatyczne aktualizacje.

6. Proszę sprawdzić czy w systemie możliwe jest logowanie na użytkownika „root”

```
wesoly@Ubuntu:~$ sudo passwd -S root
root L 2025-08-05 0 99999 7 -1
```

Rysunek 7

Litera „L” oznacza, że logowanie jest zablokowane.

```
wesoly@Ubuntu:~$ ip -4 a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    altname enp2s1
    inet 192.168.222.137/24 brd 192.168.222.255 scope global dynamic noprefixroute ens33
       valid_lft 1482sec preferred_lft 1482sec
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
wesoly@Ubuntu:~$ s^C
wesoly@Ubuntu:~$ ip -6 a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 fe80::20c:29ff:fea2:32e0/64 scope link
       valid_lft forever preferred_lft forever
wesoly@Ubuntu:~$ ^C
wesoly@Ubuntu:~$ ip route
default via 192.168.222.2 dev ens33 proto dhcp src 192.168.222.137 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.222.0/24 dev ens33 proto kernel scope link src 192.168.222.137 metric 100
wesoly@Ubuntu:~$ ip -6 route
fe80::/64 dev ens33 proto kernel metric 256 pref medium
```

Rysunek 8 Skonfigurowane adresy

Brak rozbieżności.

Zadanie 2

```
Details:
Hardening index : 63 [############        ]
Tests performed : 271
Plugins enabled : 0
```

Rysunek 9 Wynik programu Lynis przed hardeningiem

```
Details:
Hardening index : 90 [##################  ]
Tests performed : 262
Plugins enabled : 0
```

Rysunek 10 Wynik programu Lynis po hardeningu

Zadanie 3

```
Details:
Hardening index : 67 [#############        ]
Tests performed : 256
Plugins enabled : 0
```

Rysunek 11 Skan programem Lynis

```
root@Ubuntu:/home/wesoly/Downloads/scap/build/guides# ls
ssg-ubuntu2404-guide-cis_level1_server.html        ssg-ubuntu2404-guide-cis_level2_workstation.html
ssg-ubuntu2404-guide-cis_level1_workstation.html   ssg-ubuntu2404-guide-index.html
ssg-ubuntu2404-guide-cis_level2_server.html        ssg-ubuntu2404-guide-stig.html
```

Rysunek 12 Folder z pobranymi profilami

```
root@Ubuntu:/home/wesoly/Downloads/scap/build/guides# oscap xccdf eval --profile xccdf_org.ssgproject.content_profile_cis_level1_server --report raport.html --results wyniki.xml ../ss
g-ubuntu2404-ds.xml
--- Starting Evaluation ---

Title   Package "prelink" Must not be Installed
Rule    xccdf_org.ssgproject.content_rule_package_prelink_removed
Result  pass

Title   Install AIDE
Rule    xccdf_org.ssgproject.content_rule_package_aide_installed
Result  fail

Title   Build and Test AIDE Database
Rule    xccdf_org.ssgproject.content_rule_aide_build_database
Result  fail

Title   Configure Systemd Timer Execution of AIDE
Rule    xccdf_org.ssgproject.content_rule_aide_periodic_checking_systemd_timer
Result  notapplicable

Title   Ensure /dev/shm is configured
```

Rysunek 13 Proces wykonywania testu

Compliance and Scoring

The target system did not satisfy the conditions of 142 rules! Please review rule results and consider applying remediation.

Rule results

224 passed | 142 failed

Severity of failed rules

8 other | 14 low | 115 medium | 5 high

Score

Scoring system	Score	Maximum	Percent
urn:xccdf:scoring:default	59.303719	100.000000	59.3%

Rysunek 14 Wynik testu

Do testu został wybrany profil *xccdf_org.ssgproject.content_profile_cis_level1_server,* ponieważ chce, aby mój zachował wysoką funkcjonalność przy jednoczesnym wysokim poziomie zabezpieczeń.

Znajdźki, które zostały naprawione:

- Wyłączenie XDMCP
- Zainstalowanie i skonfigurowanie AIDE
- Nadanie hasła do bootloadera grub2
- Wyłączenie możliwości logowania na roota przez ssh
- Usunięcie informacji o systemie z /etc/issue*
- Dodanie polityk haseł w PAM
- Zmiana uprawnień do logów
- Zmiana wartości w zmiennych jądra

```
Details:
Hardening index : 75 [###############     ]
Tests performed : 262
Plugins enabled : 0
```

Rysunek 15 Ponowny skan Lynisem

```
root@Ubuntu:/home/wesoly/Downloads/scap/build/guides# oscap xccdf generate fix --fix-type bash --profile xccdf_org.ssgproject.content_profile_cis_level1_server --output fix.sh wyniki.
xml
root@Ubuntu:/home/wesoly/Downloads/scap/build/guides# ./fix.sh
Remediating rule 1/408: 'xccdf_org.ssgproject.content_rule_package_prelink_removed'
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'prelink' is not installed, so not removed
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Remediating rule 2/408: 'xccdf_org.ssgproject.content_rule_package_aide_installed'
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
aide is already the newest version (0.18.6-2ubuntu0.1).
```

Rysunek 16 Generowanie skryptu i wykonanie

```
  Details:
  Hardening index : 81 [################    ]
  Tests performed : 262
  Plugins enabled : 0

  Software components:
  - Firewall               [V]
  - Intrusion software     [X]
  - Malware scanner        [X]

  Files:
  - Test and debug information      : /var/log/lynis.log
  - Report data                     : /var/log/lynis-report.dat

================================================================================

  Lynis 3.1.6

  Auditing, system hardening, and compliance for UNIX-based systems
  (Linux, macOS, BSD, and others)

  2007-2025, CISOfy - https://cisofy.com/lynis/
  Enterprise support available (compliance, plugins, interface and tools)

================================================================================

  [TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)
```

Rysunek 17 Końcowy raport lynisa

```
root@Ubuntu:/home/wesoly# lynis audit system | tee a lynis-raport.txt

[ Lynis 3.1.6 ]

################################################################################
  Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
  welcome to redistribute it under the terms of the GNU General Public License.
  See the LICENSE file for details about using this software.

  2007-2025, CISOfy - https://cisofy.com/lynis/
  Enterprise support available (compliance, plugins, interface and tools)
################################################################################


[+] Initializing program
------------------------------------
  - Detecting OS...                                           [ DONE ]
  - Checking profiles...                                      [ DONE ]

  ---------------------------------------------------
  Program version:           3.1.6
  Operating system:          Linux
  Operating system name:     Ubuntu
  Operating system version:  24.04
  End-of-life:               NO
  Kernel version:            6.14.0
  Hardware platform:         x86_64
  Hostname:                  Ubuntu
  ---------------------------------------------------
  Profiles:                  /etc/lynis/default.prf
  Log file:                  /var/log/lynis.log
  Report file:               /var/log/lynis-report.dat
  Report version:            1.0
  Plugin directory:          /usr/share/lynis/plugins
  ---------------------------------------------------
  Auditor:                   [Not Specified]
  Language:                  en
  Test category:             all
  Test group:                all
  ---------------------------------------------------
  - Program update status...                                  [ NO UPDATE ]

[+] System tools
```

```
------------------------------------
  - Scanning available tools...
  - Checking system binaries...

[+] Plugins (phase 1)
------------------------------------
 Note: plugins have more extensive tests and may take several minutes to complete

  - Plugins enabled                                   [ NONE ]

[+] Boot and services
------------------------------------
  - Service Manager                                   [ systemd ]
  - Checking UEFI boot                                [ DISABLED ]
  - Checking presence GRUB2                           [ FOUND ]
    - Checking for password protection                [ NONE ]
  - Check running services (systemctl)                [ DONE ]
       Result: found 39 running services
  - Check enabled services at boot (systemctl)        [ DONE ]
       Result: found 62 enabled services
  - Check startup files (permissions)                 [ OK ]
  - Running 'systemd-analyze security'
     Unit name (exposure value) and predicate
     --------------------------------
    - ModemManager.service (value=6.3)                [ MEDIUM ]
    - NetworkManager.service (value=7.8)              [ EXPOSED ]
    - accounts-daemon.service (value=5.5)             [ MEDIUM ]
    - alsa-state.service (value=9.6)                  [ UNSAFE ]
    - anacron.service (value=9.6)                     [ UNSAFE ]
    - apache2.service (value=9.2)                     [ UNSAFE ]
    - avahi-daemon.service (value=9.6)                [ UNSAFE ]
    - colord.service (value=3.5)                      [ PROTECTED ]
    - containerd.service (value=9.6)                  [ UNSAFE ]
    - cron.service (value=9.6)                        [ UNSAFE ]
    - cups-browsed.service (value=9.3)                [ UNSAFE ]
    - cups.service (value=9.6)                        [ UNSAFE ]
    - dbus.service (value=9.5)                        [ UNSAFE ]
    - dmesg.service (value=9.6)                       [ UNSAFE ]
    - docker.service (value=9.6)                      [ UNSAFE ]
    - emergency.service (value=9.5)                   [ UNSAFE ]
    - fwupd.service (value=7.7)                       [ EXPOSED ]
    - gdm.service (value=9.8)                         [ UNSAFE ]
    - getty@tty1.service (value=9.6)                  [ UNSAFE ]
    - gnome-remote-desktop.service (value=9.2)        [ UNSAFE ]
    - kerneloops.service (value=9.2)                  [ UNSAFE ]
    - lynis.service (value=9.6)                       [ UNSAFE ]
    - mysql.service (value=9.2)                       [ UNSAFE ]
    - networkd-dispatcher.service (value=9.6)         [ UNSAFE ]
    - ntpsec-rotate-stats.service (value=9.6)         [ UNSAFE ]
    - ntpsec-systemd-netif.service (value=9.6)        [ UNSAFE ]
    - ntpsec.service (value=9.2)                      [ UNSAFE ]
    - open-vm-tools.service (value=9.5)               [ UNSAFE ]
    - packagekit.service (value=9.6)                  [ UNSAFE ]
```

```
  - plymouth-start.service (value=9.5)                    [ UNSAFE ]
  - polkit.service (value=1.6)                            [ PROTECTED ]
  - postgresql@16-main.service (value=9.6)                [ UNSAFE ]
  - power-profiles-daemon.service (value=5.1)             [ MEDIUM ]
  - rc-local.service (value=9.6)                          [ UNSAFE ]
  - rescue.service (value=9.5)                            [ UNSAFE ]
  - rsyslog.service (value=6.3)                           [ MEDIUM ]
  - rtkit-daemon.service (value=7.2)                      [ MEDIUM ]
  - snapd.service (value=9.9)                             [ UNSAFE ]
  - ssh.service (value=9.6)                               [ UNSAFE ]
  - sssd-autofs.service (value=9.6)                       [ UNSAFE ]
  - sssd-nss.service (value=9.6)                          [ UNSAFE ]
  - sssd-pac.service (value=9.6)                          [ UNSAFE ]
  - sssd-pam.service (value=9.6)                          [ UNSAFE ]
  - sssd-ssh.service (value=9.6)                          [ UNSAFE ]
  - sssd-sudo.service (value=9.6)                         [ UNSAFE ]
  - sssd.service (value=8.3)                              [ EXPOSED ]
  - switcheroo-control.service (value=7.6)                [ EXPOSED ]
  - systemd-ask-password-console.service (value=9.4)      [ UNSAFE ]
  - systemd-ask-password-plymouth.service (value=9.5)     [ UNSAFE ]
  - systemd-ask-password-wall.service (value=9.4)         [ UNSAFE ]
  - systemd-bsod.service (value=9.5)                      [ UNSAFE ]
  - systemd-fsckd.service (value=9.5)                     [ UNSAFE ]
  - systemd-initctl.service (value=9.4)                   [ UNSAFE ]
  - systemd-journald.service (value=4.3)                  [ PROTECTED ]
  - systemd-logind.service (value=2.8)                    [ PROTECTED ]
  - systemd-networkd.service (value=2.6)                  [ PROTECTED ]
  - systemd-oomd.service (value=1.8)                      [ PROTECTED ]
  - systemd-resolved.service (value=2.2)                  [ PROTECTED ]
  - systemd-rfkill.service (value=9.4)                    [ UNSAFE ]
  - systemd-udevd.service (value=7.1)                     [ MEDIUM ]
  - thermald.service (value=9.6)                          [ UNSAFE ]
  - tpm-udev.service (value=9.6)                          [ UNSAFE ]
  - ubuntu-advantage.service (value=9.6)                  [ UNSAFE ]
  - udisks2.service (value=9.6)                           [ UNSAFE ]
  - unattended-upgrades.service (value=9.6)               [ UNSAFE ]
  - upower.service (value=2.4)                            [ PROTECTED ]
  - user@1000.service (value=9.4)                         [ UNSAFE ]
  - uuidd.service (value=5.8)                             [ MEDIUM ]
  - vgauth.service (value=9.5)                            [ UNSAFE ]
  - whoopsie.service (value=9.6)                          [ UNSAFE ]
  - wpa_supplicant.service (value=9.6)                    [ UNSAFE ]

[+] Kernel
------------------------------------
  - Checking default runlevel                             [ runlevel 5 ]
  - Checking CPU support (NX/PAE)
    CPU support: PAE and/or NoeXecute supported           [ FOUND ]
  - Checking kernel version and release                   [ DONE ]
  - Checking kernel type                                  [ DONE ]
  - Checking loaded kernel modules                        [ DONE ]
      Found 101 active modules
  - Checking Linux kernel configuration file              [ FOUND ]
```

```
  - Checking default I/O kernel scheduler                   [ NOT FOUND ]
  - Checking for available kernel update                    [ OK ]
  - Checking core dumps configuration
    - configuration in systemd conf files                   [ DEFAULT ]
    - configuration in /etc/profile                         [ DEFAULT ]
    - 'hard' configuration in /etc/security/limits.conf     [ DEFAULT ]
    - 'soft' configuration in /etc/security/limits.conf     [ DEFAULT ]
    - Checking setuid core dumps configuration              [ PROTECTED ]
  - Check if reboot is needed                               [ NO ]

[+] Memory and Processes
------------------------------------
  - Checking /proc/meminfo                                  [ FOUND ]
  - Searching for dead/zombie processes                     [ NOT FOUND ]
  - Searching for IO waiting processes                      [ NOT FOUND ]
  - Search prelink tooling                                  [ NOT FOUND ]

[+] Users, Groups and Authentication
------------------------------------
  - Administrator accounts                                  [ OK ]
  - Unique UIDs                                             [ OK ]
  - Consistency of group files (grpck)                      [ OK ]
  - Unique group IDs                                        [ OK ]
  - Unique group names                                      [ OK ]
  - Password file consistency                               [ OK ]
  - Password hashing methods                                [ SUGGESTION ]
  - Checking password hashing rounds                        [ DISABLED ]
  - Query system users (non daemons)                        [ DONE ]
  - NIS+ authentication support                             [ NOT ENABLED ]
  - NIS authentication support                              [ NOT ENABLED ]
  - Sudoers file(s)                                         [ FOUND ]
    - Permissions for directory: /etc/sudoers.d             [ WARNING ]
    - Permissions for: /etc/sudoers                         [ OK ]
    - Permissions for: /etc/sudoers.d/README                [ OK ]
  - PAM password strength tools                             [ OK ]
  - PAM configuration files (pam.conf)                      [ FOUND ]
  - PAM configuration files (pam.d)                         [ FOUND ]
  - PAM modules                                             [ FOUND ]
  - LDAP module in PAM                                      [ NOT FOUND ]
  - Accounts without expire date                            [ SUGGESTION ]
  - Accounts without password                               [ OK ]
  - Locked accounts                                         [ OK ]
  - User password aging (minimum)                           [ CONFIGURED ]
  - User password aging (maximum)                           [ CONFIGURED ]
  - Checking expired passwords                              [ OK ]
  - Checking Linux single user mode authentication          [ OK ]
  - Determining default umask
    - umask (/etc/profile)                                  [ NOT FOUND ]
    - umask (/etc/login.defs)                               [ SUGGESTION ]
  - LDAP authentication support                             [ NOT ENABLED ]
  - Logging failed login attempts                           [ ENABLED ]

[+] Kerberos
```

```
------------------------------------
  - Check for Kerberos KDC and principals                    [ NOT FOUND ]

[+] Shells
------------------------------------
  - Checking shells from /etc/shells
    Result: found 7 shells (valid shells: 7).
    - Session timeout settings/tools                         [ NONE ]
  - Checking default umask values
    - Checking default umask in /etc/bash.bashrc             [ NONE ]
    - Checking default umask in /etc/profile                 [ NONE ]

[+] File systems
------------------------------------
  - Checking mount points
    - Checking /home mount point                             [ SUGGESTION ]
    - Checking /tmp mount point                              [ SUGGESTION ]
    - Checking /var mount point                              [ SUGGESTION ]
  - Query swap partitions (fstab)                            [ OK ]
  - Testing swap partitions                                  [ OK ]
  - Testing /proc mount (hidepid)                            [ SUGGESTION ]
  - Checking for old files in /tmp                           [ OK ]
  - Checking /tmp sticky bit                                 [ OK ]
  - Checking /var/tmp sticky bit                             [ OK ]
  - ACL support root file system                             [ ENABLED ]
  - Mount options of /                                       [ OK ]
  - Mount options of /dev                                    [ PARTIALLY HARDENED ]
  - Mount options of /dev/shm                                [ PARTIALLY HARDENED ]
  - Mount options of /run                                    [ HARDENED ]
  - Total without nodev:8 noexec:23 nosuid:17 ro or noexec (W^X): 12 of total 39
  - Disable kernel support of some filesystems

[+] USB Devices
------------------------------------
  - Checking usb-storage driver (modprobe config)            [ NOT DISABLED ]
  - Checking USB devices authorization                       [ ENABLED ]
  - Checking USBGuard                                        [ NOT FOUND ]

[+] Storage
------------------------------------
  - Checking firewire ohci driver (modprobe config)          [ DISABLED ]

[+] NFS
------------------------------------
  - Check running NFS daemon                                 [ NOT FOUND ]

[+] Name services
------------------------------------
  - Checking search domains                                  [ FOUND ]
  - Checking /etc/resolv.conf options                        [ FOUND ]
  - Searching DNS domain name                                [ UNKNOWN ]
  - Checking /etc/hosts
    - Duplicate entries in hosts file                        [ NONE ]
```

```
   - Presence of configured hostname in /etc/hosts          [ FOUND ]
   - Hostname mapped to localhost                           [ NOT FOUND ]
   - Localhost mapping to IP address                        [ OK ]

[+] Ports and packages
------------------------------------
  - Searching package managers
    - Searching dpkg package manager                        [ FOUND ]
      - Querying package manager
    - Query unpurged packages                               [ FOUND ]
  - Checking security repository in sources.list.d directory  [ OK ]
  - Checking APT package database                           [ OK ]
W: Target Packages (stable/binary-amd64/Packages) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target Packages (stable/binary-all/Packages) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target Translations (stable/i18n/Translation-en) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11 (stable/dep11/Components-amd64.yml) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11 (stable/dep11/Components-all.yml) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11-icons-small (stable/dep11/icons-48x48.tar) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11-icons (stable/dep11/icons-64x64.tar) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11-icons-hidpi (stable/dep11/icons-64x64@2.tar) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target CNF (stable/cnf/Commands-amd64) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target CNF (stable/cnf/Commands-all) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target Packages (stable/binary-amd64/Packages) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target Packages (stable/binary-all/Packages) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target Translations (stable/i18n/Translation-en) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11 (stable/dep11/Components-amd64.yml) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11 (stable/dep11/Components-all.yml) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11-icons-small (stable/dep11/icons-48x48.tar) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11-icons (stable/dep11/icons-64x64.tar) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target DEP-11-icons-hidpi (stable/dep11/icons-64x64@2.tar) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target CNF (stable/cnf/Commands-amd64) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
W: Target CNF (stable/cnf/Commands-all) is configured multiple times in
/etc/apt/sources.list.d/docker.list:1 and /etc/apt/sources.list.d/docker.sources:1
  - Checking vulnerable packages                            [ WARNING ]
```

```
  [WARNING]: Test PKGS-7392 had a long execution: 116.104070 seconds

  - Checking upgradeable packages                    [ SKIPPED ]
  - Checking package audit tool                      [ INSTALLED ]
    Found: apt-get
  - Toolkit for automatic upgrades (unattended-upgrade)      [ FOUND ]

[+] Networking
------------------------------------
  - Checking IPv6 configuration                      [ ENABLED ]
      Configuration method                           [ AUTO ]
      IPv6 only                                      [ NO ]
  - Checking configured nameservers
    - Testing nameservers
        Nameserver: 127.0.0.53                       [ OK ]
    - DNSSEC supported (systemd-resolved)            [ UNKNOWN ]
  - Getting listening ports (TCP/UDP)                [ DONE ]
  - Checking promiscuous interfaces                  [ OK ]
  - Checking status DHCP client                      [ NOT ACTIVE ]
  - Checking for ARP monitoring software             [ NOT FOUND ]
  - Uncommon network protocols                       [ 0 ]

[+] Printers and Spools
------------------------------------
  - Checking cups daemon                             [ RUNNING ]
  - Checking CUPS configuration file                 [ OK ]
    - File permissions                               [ WARNING ]
  - Checking CUPS addresses/sockets                  [ FOUND ]
  - Checking lp daemon                               [ NOT RUNNING ]

[+] Software: e-mail and messaging
------------------------------------

[+] Software: firewalls
------------------------------------
  - Checking iptables kernel module                  [ FOUND ]
    - Checking iptables policies of chains           [ FOUND ]
      - Chain INPUT (table: filter, target: DROP)    [ DROP ]
      - Chain INPUT (table: security, target: ACCEPT)      [ ACCEPT ]
    - Checking for empty ruleset                     [ OK ]
    - Checking for unused rules                      [ FOUND ]
  - Checking host based firewall                     [ ACTIVE ]

[+] Software: webserver
------------------------------------
  - Checking Apache (binary /usr/sbin/apache2)       [ FOUND ]
      Info: Configuration file found (/etc/apache2/apache2.conf)
      Info: No virtual hosts found
    * Loadable modules                               [ FOUND (118) ]
        - Found 118 loadable modules
          mod_evasive: anti-DoS/brute force          [ NOT FOUND ]
          mod_reqtimeout/mod_qos                     [ FOUND ]
```

```
    ModSecurity: web application firewall          [ NOT FOUND ]
- Checking TraceEnable setting in:
  /etc/apache2/mods-enabled/alias.conf              [ NOT FOUND ]
  /etc/apache2/mods-enabled/mpm_event.conf          [ NOT FOUND ]
  /etc/apache2/mods-enabled/deflate.conf            [ NOT FOUND ]
  /etc/apache2/mods-enabled/status.conf             [ NOT FOUND ]
  /etc/apache2/mods-enabled/negotiation.conf        [ NOT FOUND ]
  /etc/apache2/mods-enabled/setenvif.conf           [ NOT FOUND ]
  /etc/apache2/mods-enabled/mime.conf               [ NOT FOUND ]
  /etc/apache2/mods-enabled/reqtimeout.conf         [ NOT FOUND ]
  /etc/apache2/mods-enabled/dir.conf                [ NOT FOUND ]
  /etc/apache2/mods-enabled/autoindex.conf          [ NOT FOUND ]
  /etc/apache2/mods-available/alias.conf            [ NOT FOUND ]
  /etc/apache2/mods-available/mpm_event.conf        [ NOT FOUND ]
  /etc/apache2/mods-available/proxy_ftp.conf        [ NOT FOUND ]
  /etc/apache2/mods-available/mpm_worker.conf       [ NOT FOUND ]
  /etc/apache2/mods-available/dav_fs.conf           [ NOT FOUND ]
  /etc/apache2/mods-available/proxy_balancer.conf   [ NOT FOUND ]
  /etc/apache2/mods-available/deflate.conf          [ NOT FOUND ]
  /etc/apache2/mods-available/ssl.conf              [ NOT FOUND ]
  /etc/apache2/mods-available/ldap.conf             [ NOT FOUND ]
  /etc/apache2/mods-available/status.conf           [ NOT FOUND ]
  /etc/apache2/mods-available/negotiation.conf      [ NOT FOUND ]
  /etc/apache2/mods-available/actions.conf          [ NOT FOUND ]
  /etc/apache2/mods-available/setenvif.conf         [ NOT FOUND ]
  /etc/apache2/mods-available/mime_magic.conf       [ NOT FOUND ]
  /etc/apache2/mods-available/http2.conf            [ NOT FOUND ]
  /etc/apache2/mods-available/cgid.conf             [ NOT FOUND ]
  /etc/apache2/mods-available/cache_disk.conf       [ NOT FOUND ]
  /etc/apache2/mods-available/proxy_html.conf       [ NOT FOUND ]
  /etc/apache2/mods-available/mime.conf             [ NOT FOUND ]
  /etc/apache2/mods-available/proxy.conf            [ NOT FOUND ]
  /etc/apache2/mods-available/reqtimeout.conf       [ NOT FOUND ]
  /etc/apache2/mods-available/dir.conf              [ NOT FOUND ]
  /etc/apache2/mods-available/userdir.conf          [ NOT FOUND ]
  /etc/apache2/mods-available/autoindex.conf        [ NOT FOUND ]
  /etc/apache2/mods-available/info.conf             [ NOT FOUND ]
  /etc/apache2/mods-available/mpm_prefork.conf      [ NOT FOUND ]
  /etc/apache2/conf-enabled/serve-cgi-bin.conf      [ NOT FOUND ]
  /etc/apache2/conf-enabled/localized-error-pages.conf   [ NOT FOUND ]
  /etc/apache2/conf-enabled/charset.conf            [ NOT FOUND ]
  /etc/apache2/conf-enabled/security.conf           [ FOUND ]
  /etc/apache2/conf-enabled/other-vhosts-access-log.conf  [ NOT FOUND ]
  /etc/apache2/sites-enabled/000-default.conf       [ NOT FOUND ]
  /etc/apache2/apache2.conf                         [ NOT FOUND ]
  /etc/apache2/sites-available/default-ssl.conf     [ NOT FOUND ]
  /etc/apache2/sites-available/000-default.conf     [ NOT FOUND ]
  /etc/apache2/conf-available/serve-cgi-bin.conf    [ NOT FOUND ]
  /etc/apache2/conf-available/localized-error-pages.conf  [ NOT FOUND ]
  /etc/apache2/conf-available/charset.conf          [ NOT FOUND ]
  /etc/apache2/conf-available/security.conf         [ FOUND ]
  /etc/apache2/conf-available/other-vhosts-access-log.conf  [ NOT FOUND ]
  /etc/apache2/ports.conf                           [ NOT FOUND ]
```

```
  - Checking nginx                                    [ NOT FOUND ]

[+] SSH Support
------------------------------------
  - Checking running SSH daemon                       [ FOUND ]
    - Searching SSH configuration                     [ FOUND ]
    - OpenSSH option: AllowTcpForwarding              [ SUGGESTION ]
    - OpenSSH option: ClientAliveCountMax             [ SUGGESTION ]
    - OpenSSH option: ClientAliveInterval             [ OK ]
    - OpenSSH option: FingerprintHash                 [ OK ]
    - OpenSSH option: GatewayPorts                    [ OK ]
    - OpenSSH option: IgnoreRhosts                    [ OK ]
    - OpenSSH option: LoginGraceTime                  [ OK ]
    - OpenSSH option: LogLevel                        [ SUGGESTION ]
    - OpenSSH option: MaxAuthTries                    [ SUGGESTION ]
    - OpenSSH option: MaxSessions                     [ SUGGESTION ]
    - OpenSSH option: PermitRootLogin                 [ OK ]
    - OpenSSH option: PermitUserEnvironment           [ OK ]
    - OpenSSH option: PermitTunnel                    [ OK ]
    - OpenSSH option: Port                            [ SUGGESTION ]
    - OpenSSH option: PrintLastLog                    [ OK ]
    - OpenSSH option: StrictModes                     [ OK ]
    - OpenSSH option: TCPKeepAlive                    [ SUGGESTION ]
    - OpenSSH option: UseDNS                          [ OK ]
    - OpenSSH option: X11Forwarding                   [ SUGGESTION ]
    - OpenSSH option: AllowAgentForwarding            [ SUGGESTION ]
    - OpenSSH option: AllowUsers                      [ NOT FOUND ]
    - OpenSSH option: AllowGroups                     [ NOT FOUND ]

[+] SNMP Support
------------------------------------
  - Checking running SNMP daemon                      [ NOT FOUND ]

[+] Databases
------------------------------------
  - MySQL process status                              [ FOUND ]
  - PostgreSQL processes status                       [ FOUND ]

[+] LDAP Services
------------------------------------
  - Checking OpenLDAP instance                        [ NOT FOUND ]

[+] PHP
------------------------------------
  - Checking PHP                                      [ NOT FOUND ]

[+] Squid Support
------------------------------------
  - Checking running Squid daemon                     [ NOT FOUND ]

[+] Logging and files
------------------------------------
  - Checking for a running log daemon                 [ OK ]
```

```
    - Checking Syslog-NG status                     [ NOT FOUND ]
    - Checking systemd journal status               [ FOUND ]
    - Checking Metalog status                       [ NOT FOUND ]
    - Checking RSyslog status                       [ FOUND ]
    - Checking RFC 3195 daemon status               [ NOT FOUND ]
    - Checking minilogd instances                   [ NOT FOUND ]
    - Checking wazuh-agent daemon status            [ NOT FOUND ]
  - Checking logrotate presence                     [ OK ]
  - Checking remote logging                         [ NOT ENABLED ]
  - Checking log directories (static list)          [ DONE ]
  - Checking open log files                         [ DONE ]
  - Checking deleted files in use                   [ FILES FOUND ]

[+] Insecure services
------------------------------------
  - Installed inetd package                         [ NOT FOUND ]
  - Installed xinetd package                        [ OK ]
    - xinetd status                                 [ NOT ACTIVE ]
  - Installed rsh client package                    [ OK ]
  - Installed rsh server package                    [ OK ]
  - Installed telnet client package                 [ OK ]
  - Installed telnet server package                 [ NOT FOUND ]
  - Checking NIS client installation                [ OK ]
  - Checking NIS server installation                [ OK ]
  - Checking TFTP client installation               [ OK ]
  - Checking TFTP server installation               [ OK ]

[+] Banners and identification
------------------------------------
  - /etc/issue                                      [ FOUND ]
    - /etc/issue contents                           [ WEAK ]
  - /etc/issue.net                                  [ FOUND ]
    - /etc/issue.net contents                       [ WEAK ]

[+] Scheduled tasks
------------------------------------
  - Checking crontab and cronjob files              [ DONE ]

[+] Accounting
------------------------------------
  - Checking accounting information                 [ NOT FOUND ]
  - Checking sysstat accounting data                [ DISABLED ]
  - Checking auditd                                 [ NOT FOUND ]

[+] Time and Synchronization
------------------------------------
  - NTP daemon found: ntpd                          [ FOUND ]
  - Checking for a running NTP daemon or client     [ OK ]
  - Checking valid association ID's                 [ FOUND ]
  - Checking high stratum ntp peers                 [ OK ]
  - Checking unreliable ntp peers                   [ FOUND ]
  - Checking selected time source                   [ OK ]
  - Checking time source candidates                 [ OK ]
```

```
  - Checking falsetickers                          [ OK ]
  - Checking NTP version                           [ FOUND ]

[+] Cryptography
------------------------------------
  - Checking for expired SSL certificates [0/151]          [ NONE ]

  [WARNING]: Test CRYP-7902 had a long execution: 21.352726 seconds

  - Kernel entropy is sufficient                   [ YES ]
  - HW RNG & rngd                                  [ NO ]
  - SW prng                                        [ NO ]
  - MOR variable not found                         [ WEAK ]

[+] Virtualization
------------------------------------

[+] Containers
------------------------------------
    - Docker
      - Docker daemon                              [ RUNNING ]
        - Docker info output (warnings)            [ NONE ]
      - Containers
        - Total containers                         [ 0 ]
    - File permissions                             [ OK ]

[+] Security frameworks
------------------------------------
  - Checking presence AppArmor                     [ FOUND ]
    - Checking AppArmor status                     [ ENABLED ]
        Found 130 unconfined processes
  - Checking presence SELinux                      [ NOT FOUND ]
  - Checking presence TOMOYO Linux                 [ NOT FOUND ]
  - Checking presence grsecurity                   [ NOT FOUND ]
  - Checking for implemented MAC framework         [ OK ]

[+] Software: file integrity
------------------------------------
  - Checking file integrity tools
  - Checking presence integrity tool               [ NOT FOUND ]

[+] Software: System tooling
------------------------------------
  - Checking automation tooling
  - Automation tooling                             [ NOT FOUND ]
  - Checking for IDS/IPS tooling                   [ NONE ]

[+] Software: Malware
------------------------------------
  - Malware software components                     [ NOT FOUND ]

[+] File Permissions
------------------------------------
```

```
  - Starting file permissions check
    File: /boot/grub/grub.cfg                          [ OK ]
    File: /etc/crontab                                 [ SUGGESTION ]
    File: /etc/group                                   [ OK ]
    File: /etc/group-                                  [ OK ]
    File: /etc/hosts.allow                             [ OK ]
    File: /etc/hosts.deny                              [ OK ]
    File: /etc/issue                                   [ OK ]
    File: /etc/issue.net                               [ OK ]
    File: /etc/passwd                                  [ OK ]
    File: /etc/passwd-                                 [ OK ]
    File: /etc/ssh/sshd_config                         [ SUGGESTION ]
    Directory: /root/.ssh                              [ OK ]
    Directory: /etc/cron.d                             [ SUGGESTION ]
    Directory: /etc/cron.daily                         [ SUGGESTION ]
    Directory: /etc/cron.hourly                        [ SUGGESTION ]
    Directory: /etc/cron.weekly                        [ SUGGESTION ]
    Directory: /etc/cron.monthly                       [ SUGGESTION ]

[+] Home directories
------------------------------------
  - Permissions of home directories                    [ WARNING ]
  - Ownership of home directories                      [ OK ]
  - Checking shell history files                       [ OK ]

[+] Kernel Hardening
------------------------------------
  - Comparing sysctl key pairs with scan profile
    - dev.tty.ldisc_autoload (exp: 0)                  [ DIFFERENT ]
    - fs.protected_fifos (exp: 2)                      [ DIFFERENT ]
    - fs.protected_hardlinks (exp: 1)                  [ OK ]
    - fs.protected_regular (exp: 2)                    [ OK ]
    - fs.protected_symlinks (exp: 1)                   [ OK ]
    - fs.suid_dumpable (exp: 0)                        [ DIFFERENT ]
    - kernel.core_uses_pid (exp: 1)                    [ DIFFERENT ]
    - kernel.ctrl-alt-del (exp: 0)                     [ OK ]
    - kernel.dmesg_restrict (exp: 1)                   [ OK ]
    - kernel.kptr_restrict (exp: 2)                    [ DIFFERENT ]
    - kernel.modules_disabled (exp: 1)                 [ DIFFERENT ]
    - kernel.perf_event_paranoid (exp: 2 3 4)          [ OK ]
    - kernel.randomize_va_space (exp: 2)               [ OK ]
    - kernel.sysrq (exp: 0)                            [ DIFFERENT ]
    - kernel.unprivileged_bpf_disabled (exp: 1)        [ DIFFERENT ]
    - kernel.yama.ptrace_scope (exp: 1 2 3)            [ OK ]
    - net.core.bpf_jit_harden (exp: 2)                 [ DIFFERENT ]
    - net.ipv4.conf.all.accept_redirects (exp: 0)      [ OK ]
    - net.ipv4.conf.all.accept_source_route (exp: 0)   [ OK ]
    - net.ipv4.conf.all.bootp_relay (exp: 0)           [ OK ]
    - net.ipv4.conf.all.forwarding (exp: 0)            [ DIFFERENT ]
    - net.ipv4.conf.all.log_martians (exp: 1)          [ DIFFERENT ]
    - net.ipv4.conf.all.mc_forwarding (exp: 0)         [ OK ]
    - net.ipv4.conf.all.proxy_arp (exp: 0)             [ OK ]
    - net.ipv4.conf.all.rp_filter (exp: 1)             [ DIFFERENT ]
```

```
  - net.ipv4.conf.all.send_redirects (exp: 0)             [ DIFFERENT ]
  - net.ipv4.conf.default.accept_redirects (exp: 0)        [ OK ]
  - net.ipv4.conf.default.accept_source_route (exp: 0)      [ DIFFERENT ]
  - net.ipv4.conf.default.log_martians (exp: 1)            [ DIFFERENT ]
  - net.ipv4.icmp_echo_ignore_broadcasts (exp: 1)          [ OK ]
  - net.ipv4.icmp_ignore_bogus_error_responses (exp: 1)     [ OK ]
  - net.ipv4.tcp_syncookies (exp: 1)                       [ OK ]
  - net.ipv4.tcp_timestamps (exp: 0 1)                     [ OK ]
  - net.ipv6.conf.all.accept_redirects (exp: 0)            [ OK ]
  - net.ipv6.conf.all.accept_source_route (exp: 0)         [ OK ]
  - net.ipv6.conf.default.accept_redirects (exp: 0)        [ OK ]
  - net.ipv6.conf.default.accept_source_route (exp: 0)      [ OK ]

[+] Hardening
------------------------------------
  - Installed compiler(s)                          [ NOT FOUND ]
  - Installed malware scanner                      [ NOT FOUND ]
  - Non-native binary formats                      [ FOUND ]

[+] Custom tests
------------------------------------
  - Running custom tests...                        [ NONE ]

[+] Plugins (phase 2)
------------------------------------

================================================================================
=======

  -[ Lynis 3.1.6 Results ]-

  Warnings (4):
  ----------------------------
  ! Found one or more vulnerable packages. [PKGS-7392]
      https://cisofy.com/lynis/controls/PKGS-7392/

  ! PostgreSQL configuration file /etc/postgresql/16/main/pg_ctl.conf is world readable and might leak
sensitive details [DBS-1828]
    - Details  : /etc/postgresql/16/main/pg_ctl.conf
    - Solution : Use chmod 600 to change file permissions
      https://cisofy.com/lynis/controls/DBS-1828/

  ! PostgreSQL configuration file /etc/postgresql/16/main/start.conf is world readable and might leak
sensitive details [DBS-1828]
    - Details  : /etc/postgresql/16/main/start.conf
    - Solution : Use chmod 600 to change file permissions
      https://cisofy.com/lynis/controls/DBS-1828/

  ! PostgreSQL configuration file /etc/postgresql/16/main/postgresql.conf is world readable and might
leak sensitive details [DBS-1828]
    - Details  : /etc/postgresql/16/main/postgresql.conf
    - Solution : Use chmod 600 to change file permissions
      https://cisofy.com/lynis/controls/DBS-1828/
```

Suggestions (47):
---
* Set a password on GRUB boot loader to prevent altering boot configuration (e.g. boot in single user mode without password) [BOOT-5122]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/BOOT-5122/

* Consider hardening system services [BOOT-5264]
  - Details : Run '/usr/bin/systemd-analyze security SERVICE' for each service
  - Related resources
    * Article: Systemd features to secure service files: https://linux-audit.com/systemd/systemd-features-to-secure-units-and-services/
    * Website: https://cisofy.com/lynis/controls/BOOT-5264/

* If not required, consider explicit disabling of core dump in /etc/security/limits.conf file [KRNL-5820]
  - Related resources
    * Article: Understand and configure core dumps on Linux: https://linux-audit.com/software/understand-and-configure-core-dumps-work-on-linux/
    * Website: https://cisofy.com/lynis/controls/KRNL-5820/

* Check PAM configuration, add rounds if applicable and expire passwords to encrypt with new values [AUTH-9229]
  - Related resources
    * Article: Linux password security: hashing rounds: https://linux-audit.com/authentication/configure-the-minimum-password-length-on-linux-systems/
    * Website: https://cisofy.com/lynis/controls/AUTH-9229/

* Configure password hashing rounds in /etc/login.defs [AUTH-9230]
  - Related resources
    * Article: Linux password security: hashing rounds: https://linux-audit.com/authentication/configure-the-minimum-password-length-on-linux-systems/
    * Website: https://cisofy.com/lynis/controls/AUTH-9230/

* When possible set expire dates for all password protected accounts [AUTH-9282]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/AUTH-9282/

* Default umask in /etc/login.defs could be more strict like 027 [AUTH-9328]
  - Related resources
    * Article: Set default file permissions on Linux with umask: https://linux-audit.com/filesystems/file-permissions/set-default-file-permissions-with-umask/
    * Website: https://cisofy.com/lynis/controls/AUTH-9328/

* To decrease the impact of a full /home file system, place /home on a separate partition [FILE-6310]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/FILE-6310/

* To decrease the impact of a full /tmp file system, place /tmp on a separate partition [FILE-6310]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/FILE-6310/

* To decrease the impact of a full /var file system, place /var on a separate partition [FILE-6310]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/FILE-6310/

* Disable drivers like USB storage when not used, to prevent unauthorized storage or data theft [USB-1000]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/USB-1000/

* Check DNS configuration for the dns domain name [NAME-4028]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/NAME-4028/

* Purge old/removed packages (2 found) with aptitude purge or dpkg --purge command. This will cleanup old configuration files, cron jobs and startup scripts. [PKGS-7346]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/PKGS-7346/

* Install debsums utility for the verification of packages with known good database. [PKGS-7370]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/PKGS-7370/

* Update your system with apt-get update, apt-get upgrade, apt-get dist-upgrade and/or unattended-upgrades [PKGS-7392]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/PKGS-7392/

* Install package apt-show-versions for patch management purposes [PKGS-7394]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/PKGS-7394/

* Determine if protocol 'dccp' is really needed on this system [NETW-3200]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/NETW-3200/

* Determine if protocol 'sctp' is really needed on this system [NETW-3200]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/NETW-3200/

* Determine if protocol 'rds' is really needed on this system [NETW-3200]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/NETW-3200/

* Determine if protocol 'tipc' is really needed on this system [NETW-3200]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/NETW-3200/

* Access to CUPS configuration could be more strict. [PRNT-2307]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/PRNT-2307/

* Check iptables rules to see which rules are currently not used [FIRE-4513]
  - Related resources

* Website: https://cisofy.com/lynis/controls/FIRE-4513/

* Install Apache mod_evasive to guard webserver against DoS/brute force attempts [HTTP-6640]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/HTTP-6640/

* Install Apache modsecurity to guard webserver against web application attacks [HTTP-6643]
  - Related resources
    * Website: https://cisofy.com/lynis/controls/HTTP-6643/

* Consider hardening SSH configuration [SSH-7408]
  - Details : AllowTcpForwarding (set YES to NO)
  - Related resources
    * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
    * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
  - Details : ClientAliveCountMax (set 3 to 2)
  - Related resources
    * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
    * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
  - Details : LogLevel (set INFO to VERBOSE)
  - Related resources
    * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
    * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
  - Details : MaxAuthTries (set 6 to 3)
  - Related resources
    * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
    * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
  - Details : MaxSessions (set 10 to 2)
  - Related resources
    * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
    * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
  - Details : Port (set 22 to)
  - Related resources
    * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
    * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]

- Details : TCPKeepAlive (set YES to NO)
- Related resources
  * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
  * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
- Details : X11Forwarding (set YES to NO)
- Related resources
  * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
  * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Consider hardening SSH configuration [SSH-7408]
- Details : AllowAgentForwarding (set YES to NO)
- Related resources
  * Article: OpenSSH security and hardening: https://linux-audit.com/ssh/audit-and-harden-your-ssh-configuration/
  * Website: https://cisofy.com/lynis/controls/SSH-7408/

* Enable logging to an external logging host for archiving purposes and additional protection [LOGG-2154]
- Related resources
  * Website: https://cisofy.com/lynis/controls/LOGG-2154/

* Check what deleted files are still in use and why. [LOGG-2190]
- Related resources
  * Website: https://cisofy.com/lynis/controls/LOGG-2190/

* Add a legal banner to /etc/issue, to warn unauthorized users [BANN-7126]
- Related resources
  * Article: The real purpose of login banners: https://linux-audit.com/the-real-purpose-of-login-banners-on-linux/
  * Website: https://cisofy.com/lynis/controls/BANN-7126/

* Add legal banner to /etc/issue.net, to warn unauthorized users [BANN-7130]
- Related resources
  * Article: The real purpose of login banners: https://linux-audit.com/the-real-purpose-of-login-banners-on-linux/
  * Website: https://cisofy.com/lynis/controls/BANN-7130/

* Enable process accounting [ACCT-9622]
- Related resources
  * Website: https://cisofy.com/lynis/controls/ACCT-9622/

* Enable sysstat to collect accounting (disabled) [ACCT-9626]
- Related resources
  * Website: https://cisofy.com/lynis/controls/ACCT-9626/

* Enable auditd to collect audit information [ACCT-9628]
- Related resources
  * Article: Linux audit framework 101: basic rules for configuration: https://linux-audit.com/linux-audit-framework/linux-audit-framework-101-basic-rules-for-configuration/

* Article: Monitoring Linux file access, changes and data modifications: https://linux-audit.com/monitoring-linux-file-access-changes-and-modifications/
* Website: https://cisofy.com/lynis/controls/ACCT-9628/

* Check ntpq peers output for unreliable ntp peers and correct/replace them [TIME-3120]
- Related resources
* Website: https://cisofy.com/lynis/controls/TIME-3120/

* Install a file integrity tool to monitor changes to critical and sensitive files [FINT-4350]
- Related resources
* Article: Monitoring Linux file access, changes and data modifications: https://linux-audit.com/monitoring-linux-file-access-changes-and-modifications/
* Article: Monitor for file changes on Linux: https://linux-audit.com/monitor-for-file-system-changes-on-linux/
* Website: https://cisofy.com/lynis/controls/FINT-4350/

* Determine if automation tools are present for system management [TOOL-5002]
- Related resources
* Website: https://cisofy.com/lynis/controls/TOOL-5002/

* Consider restricting file permissions [FILE-7524]
- Details : See screen output or log file
- Solution : Use chmod to change file permissions
- Related resources
* Website: https://cisofy.com/lynis/controls/FILE-7524/

* Double check the permissions of home directories as some might be not strict enough. [HOME-9304]
- Related resources
* Website: https://cisofy.com/lynis/controls/HOME-9304/

* One or more sysctl values differ from the scan profile and could be tweaked [KRNL-6000]
- Solution : Change sysctl value or disable test (skip-test=KRNL-6000:<sysctl-key>)
- Related resources
* Article: Linux hardening with sysctl settings: https://linux-audit.com/linux-hardening-with-sysctl/
* Article: Overview of sysctl options and values: https://linux-audit.com/kernel/sysctl/
* Website: https://cisofy.com/lynis/controls/KRNL-6000/

* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]
- Solution : Install a tool like rkhunter, chkrootkit, OSSEC, Wazuh
- Related resources
* Article: Antivirus for Linux: is it really needed?: https://linux-audit.com/malware/antivirus-for-linux-really-needed/
* Article: Monitoring Linux Systems for Rootkits: https://linux-audit.com/monitoring-linux-systems-for-rootkits/
* Website: https://cisofy.com/lynis/controls/HRDN-7230/

Follow-up:

- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)

```
  - Read security controls texts (https://cisofy.com)
  - Use --upload to upload data to central system (Lynis Enterprise users)

================================================================================

  Lynis security scan details:

  Scan mode:
  Normal [▇]  Forensics [ ]  Integration [ ]  Pentest [ ]

  Lynis modules:
  - Compliance status      [?]
  - Security audit         [V]
  - Vulnerability scan     [V]

  Details:
  Hardening index : 63 [############        ]
  Tests performed : 271
  Plugins enabled : 0

  Software components:
  - Firewall               [V]
  - Intrusion software     [X]
  - Malware scanner        [X]

  Files:
  - Test and debug information      : /var/log/lynis.log
  - Report data                     : /var/log/lynis-report.dat

================================================================================

  Lynis 3.1.6

  Auditing, system hardening, and compliance for UNIX-based systems
  (Linux, macOS, BSD, and others)

  2007-2025, CISOfy - https://cisofy.com/lynis/
  Enterprise support available (compliance, plugins, interface and tools)

================================================================================

  [TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all
settings)
```

```
root@ubuntu:/home/wesoly# lynis audit system | tee lynis-raport.txt

[ Lynis 3.1.6 ]

################################################################################
  Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
  welcome to redistribute it under the terms of the GNU General Public License.
  See the LICENSE file for details about using this software.

  2007-2025, CISOfy - https://cisofy.com/lynis/
  Enterprise support available (compliance, plugins, interface and tools)
################################################################################
```

```
[+] Initializing program
------------------------------------
  - Detecting OS...                                           [ DONE ]
  - Checking profiles...                                      [ DONE ]

  ---------------------------------------------------
  Program version:           3.1.6
  Operating system:          Linux
  Operating system name:     Ubuntu
  Operating system version:  24.04
  End-of-life:               NO
  Kernel version:            6.17.0
  Hardware platform:         x86_64
  Hostname:                  ubuntu
  ---------------------------------------------------
  Profiles:                  /etc/lynis/default.prf
  Log file:                  /var/log/lynis.log
  Report file:               /var/log/lynis-report.dat
  Report version:            1.0
  Plugin directory:          /usr/share/lynis/plugins
  ---------------------------------------------------
  Auditor:                   [Not Specified]
  Language:                  en
  Test category:             all
  Test group:                all
  ---------------------------------------------------
  - Program update status...                                  [ NO UPDATE ]

[+] System tools
------------------------------------
  - Scanning available tools...
  - Checking system binaries...

[+] Plugins (phase 1)
------------------------------------
 Note: plugins have more extensive tests and may take several minutes to complete

  - Plugins enabled                                           [ NONE ]

[+] Boot and services
------------------------------------
  - Service Manager                                           [ systemd ]
  - Checking UEFI boot                                        [ DISABLED ]
  - Checking presence GRUB2                                   [ FOUND ]
    - Checking for password protection                        [ OK ]
  - Check running services (systemctl)                        [ DONE ]
      Result: found 32 running services
  - Check enabled services at boot (systemctl)                [ DONE ]
      Result: found 58 enabled services
  - Check startup files (permissions)                         [ OK ]
  - Running 'systemd-analyze security'
     Unit name (exposure value) and predicate
     --------------------------------
```

- NetworkManager.service (value=7.8) [EXPOSED]
- accounts-daemon.service (value=5.5) [MEDIUM]
- alsa-state.service (value=1.5) [PROTECTED]
- anacron.service (value=1.4) [PROTECTED]
- apache2.service (value=1.4) [PROTECTED]
- auditd.service (value=8.9) [EXPOSED]
- colord.service (value=3.5) [PROTECTED]
- cron.service (value=9.6) [UNSAFE]
- cups-browsed.service (value=9.3) [UNSAFE]
- cups.service (value=0.5) [PROTECTED]
- dbus.service (value=9.5) [UNSAFE]
- dmesg.service (value=9.6) [UNSAFE]
- emergency.service (value=9.5) [UNSAFE]
- fail2ban.service (value=9.6) [UNSAFE]
- gdm.service (value=9.8) [UNSAFE]
- getty@tty1.service (value=9.6) [UNSAFE]
- gnome-remote-desktop.service (value=9.2) [UNSAFE]
- kerneloops.service (value=9.2) [UNSAFE]
- networkd-dispatcher.service (value=9.6) [UNSAFE]
- open-vm-tools.service (value=9.5) [UNSAFE]
- packagekit.service (value=9.6) [UNSAFE]
- plymouth-start.service (value=9.5) [UNSAFE]
- polkit.service (value=1.6) [PROTECTED]
- postfix@-.service (value=9.6) [UNSAFE]
- postgresql@16-main.service (value=9.6) [UNSAFE]
- power-profiles-daemon.service (value=5.1) [MEDIUM]
- rc-local.service (value=9.6) [UNSAFE]
- rescue.service (value=9.5) [UNSAFE]
- rsyslog.service (value=6.3) [MEDIUM]
- rtkit-daemon.service (value=7.2) [MEDIUM]
- snapd.service (value=9.9) [UNSAFE]
- ssh.service (value=1.5) [PROTECTED]
- systemd-ask-password-console.service (value=9.4) [UNSAFE]
- systemd-ask-password-plymouth.service (value=9.5) [UNSAFE]
- systemd-ask-password-wall.service (value=9.4) [UNSAFE]
- systemd-bsod.service (value=9.5) [UNSAFE]
- systemd-fsckd.service (value=9.5) [UNSAFE]
- systemd-initctl.service (value=9.4) [UNSAFE]
- systemd-journald.service (value=4.3) [PROTECTED]
- systemd-logind.service (value=2.8) [PROTECTED]
- systemd-networkd.service (value=2.6) [PROTECTED]
- systemd-oomd.service (value=1.8) [PROTECTED]
- systemd-resolved.service (value=2.2) [PROTECTED]
- systemd-rfkill.service (value=9.4) [UNSAFE]
- systemd-timesyncd.service (value=2.1) [PROTECTED]
- systemd-udevd.service (value=7.1) [MEDIUM]
- thermald.service (value=9.6) [UNSAFE]
- tpm-udev.service (value=9.6) [UNSAFE]
- ubuntu-advantage.service (value=9.6) [UNSAFE]
- udisks2.service (value=9.6) [UNSAFE]
- unattended-upgrades.service (value=9.6) [UNSAFE]
- upower.service (value=2.4) [PROTECTED]
- user@1000.service (value=9.4) [UNSAFE]

```
  - uuidd.service (value=5.8)                    [ MEDIUM ]
  - vgauth.service (value=9.5)                   [ UNSAFE ]
  - whoopsie.service (value=2.3)                 [ PROTECTED ]
  - wpa_supplicant.service (value=9.6)           [ UNSAFE ]

[+] Kernel
------------------------------------
 - Checking default runlevel                     [ runlevel 5 ]
 - Checking CPU support (NX/PAE)
   CPU support: PAE and/or NoeXecute supported          [ FOUND ]
 - Checking kernel version and release           [ DONE ]
 - Checking kernel type                          [ DONE ]
 - Checking loaded kernel modules                [ DONE ]
     Found 84 active modules
 - Checking Linux kernel configuration file      [ FOUND ]
 - Checking default I/O kernel scheduler         [ NOT FOUND ]
 - Checking for available kernel update          [ OK ]
 - Checking core dumps configuration
   - configuration in systemd conf files         [ DEFAULT ]
   - configuration in /etc/profile               [ DEFAULT ]
   - 'hard' configuration in /etc/security/limits.conf      [ DISABLED ]
   - 'soft' config in /etc/security/limits.conf (implicit)  [ DISABLED ]
   - Checking setuid core dumps configuration              [ DISABLED ]
 - Check if reboot is needed                     [ NO ]

[+] Memory and Processes
------------------------------------
 - Checking /proc/meminfo                        [ FOUND ]
 - Searching for dead/zombie processes           [ NOT FOUND ]
 - Searching for IO waiting processes            [ NOT FOUND ]
 - Search prelink tooling                        [ NOT FOUND ]

[+] Users, Groups and Authentication
------------------------------------
 - Administrator accounts                        [ OK ]
 - Unique UIDs                                   [ OK ]
 - Consistency of group files (grpck)            [ OK ]
 - Unique group IDs                              [ OK ]
 - Unique group names                            [ OK ]
 - Password file consistency                     [ OK ]
 - Password hashing methods                      [ OK ]
 - Password hashing rounds (minimum)             [ CONFIGURED ]
 - Query system users (non daemons)              [ DONE ]
 - NIS+ authentication support                   [ NOT ENABLED ]
 - NIS authentication support                    [ NOT ENABLED ]
 - Sudoers file(s)                               [ FOUND ]
   - Permissions for directory: /etc/sudoers.d   [ OK ]
   - Permissions for: /etc/sudoers               [ OK ]
   - Permissions for: /etc/sudoers.d/README      [ OK ]
 - PAM password strength tools                   [ OK ]
 - PAM configuration files (pam.conf)            [ FOUND ]
 - PAM configuration files (pam.d)               [ FOUND ]
 - PAM modules                                   [ FOUND ]
```

```
  - LDAP module in PAM                                    [ NOT FOUND ]
  - Accounts without expire date                          [ OK ]
  - Accounts without password                             [ OK ]
  - Locked accounts                                       [ OK ]
  - User password aging (minimum)                         [ CONFIGURED ]
  - User password aging (maximum)                         [ CONFIGURED ]
  - Checking expired passwords                            [ OK ]
  - Checking Linux single user mode authentication        [ OK ]
  - Determining default umask
    - umask (/etc/profile)                                [ NOT FOUND ]
    - umask (/etc/login.defs)                             [ OK ]
  - LDAP authentication support                           [ NOT ENABLED ]
  - Logging failed login attempts                         [ ENABLED ]

[+] Kerberos
------------------------------------
  - Check for Kerberos KDC and principals                 [ NOT FOUND ]

[+] Shells
------------------------------------
  - Checking shells from /etc/shells
    Result: found 7 shells (valid shells: 7).
    - Session timeout settings/tools                      [ NONE ]
  - Checking default umask values
    - Checking default umask in /etc/bash.bashrc          [ NONE ]
    - Checking default umask in /etc/profile              [ NONE ]

[+] File systems
------------------------------------
  - Checking mount points
    - Checking /home mount point                          [ OK ]
    - Checking /tmp mount point                           [ OK ]
    - Checking /var mount point                           [ OK ]
  - Query swap partitions (fstab)                         [ OK ]
  - Testing swap partitions                               [ OK ]
  - Testing /proc mount (hidepid)                         [ OK ]
  - Checking for old files in /tmp                        [ OK ]
  - Checking /tmp sticky bit                              [ OK ]
  - Checking /var/tmp sticky bit                          [ OK ]
  - ACL support root file system                          [ ENABLED ]
  - Mount options of /                                    [ OK ]
  - Mount options of /dev                                 [ PARTIALLY HARDENED ]
  - Mount options of /dev/shm                             [ PARTIALLY HARDENED ]
  - Mount options of /home                                [ HARDENED ]
  - Mount options of /run                                 [ HARDENED ]
  - Mount options of /tmp                                 [ HARDENED ]
  - Mount options of /var                                 [ HARDENED ]
  - Total without nodev:10 noexec:26 nosuid:18 ro or noexec (W^X): 16 of total 42
  - Disable kernel support of some filesystems

[+] USB Devices
------------------------------------
  - Checking usb-storage driver (modprobe config)         [ DISABLED ]
```

```
  - Checking USB devices authorization                    [ ENABLED ]
  - Checking USBGuard                                     [ NOT FOUND ]

[+] Storage
------------------------------------
  - Checking firewire ohci driver (modprobe config)       [ DISABLED ]

[+] NFS
------------------------------------
  - Check running NFS daemon                              [ NOT FOUND ]

[+] Name services
------------------------------------
  - Checking search domains                               [ FOUND ]
  - Checking /etc/resolv.conf options                     [ FOUND ]
  - Searching DNS domain name                             [ FOUND ]
    Domain name: home.local
  - Checking /etc/hosts
    - Duplicate entries in hosts file                     [ NONE ]
    - Presence of configured hostname in /etc/hosts       [ FOUND ]
    - Hostname mapped to localhost                        [ NOT FOUND ]
    - Localhost mapping to IP address                     [ OK ]

[+] Ports and packages
------------------------------------
  - Searching package managers
    - Searching dpkg package manager                      [ FOUND ]
      - Querying package manager
    - Query unpurged packages                             [ NONE ]
    - debsums utility                                     [ FOUND ]
      - Cron job for debsums                              [ FOUND ]
  - Checking security repository in sources.list.d directory  [ OK ]
  - Checking APT package database                         [ OK ]
  - Checking vulnerable packages                          [ OK ]

  [WARNING]: Test PKGS-7392 had a long execution: 17.954958 seconds

  - Checking upgradeable packages                         [ FOUND ]
  - Checking package audit tool                           [ INSTALLED ]
    Found: apt-check
  - Toolkit for automatic upgrades (unattended-upgrade)   [ FOUND ]

[+] Networking
------------------------------------
  - Checking IPv6 configuration                           [ ENABLED ]
      Configuration method                                [ AUTO ]
      IPv6 only                                           [ NO ]
  - Checking configured nameservers
    - Testing nameservers
        Nameserver: 127.0.0.53                            [ OK ]
    - DNSSEC supported (systemd-resolved)                 [ UNKNOWN ]
  - Checking default gateway                              [ DONE ]
  - Getting listening ports (TCP/UDP)                     [ DONE ]
```

```
  - Checking promiscuous interfaces                    [ OK ]
  - Checking waiting connections                       [ OK ]
  - Checking status DHCP client                        [ NOT ACTIVE ]
  - Checking for ARP monitoring software               [ NOT FOUND ]
  - Uncommon network protocols                         [ NOT FOUND ]

[+] Printers and Spools
------------------------------------
  - Checking cups daemon                               [ NOT FOUND ]
  - Checking lp daemon                                 [ NOT RUNNING ]

[+] Software: e-mail and messaging
------------------------------------

[+] Software: firewalls
------------------------------------
  - Checking iptables kernel module                    [ FOUND ]
    - Checking iptables policies of chains             [ FOUND ]
      - Chain INPUT (table: filter, target: DROP)      [ DROP ]
      - Chain INPUT (table: security, target: ACCEPT)  [ ACCEPT ]
    - Checking for empty ruleset                       [ OK ]
    - Checking for unused rules                        [ FOUND ]
  - Checking host based firewall                       [ ACTIVE ]

[+] Software: webserver
------------------------------------
  - Checking Apache (binary /usr/sbin/apache2)         [ FOUND ]
      Info: Configuration file found (/etc/apache2/apache2.conf)
      Info: No virtual hosts found
    * Loadable modules                                 [ FOUND (120) ]
        - Found 120 loadable modules
          mod_evasive: anti-DoS/brute force            [ FOUND ]
          mod_reqtimeout/mod_qos                       [ FOUND ]
          ModSecurity: web application firewall        [ FOUND ]
  - Checking TraceEnable setting in:
      /etc/apache2/mods-enabled/alias.conf             [ NOT FOUND ]
      /etc/apache2/mods-enabled/mpm_event.conf         [ NOT FOUND ]
      /etc/apache2/mods-enabled/deflate.conf           [ NOT FOUND ]
      /etc/apache2/mods-enabled/status.conf            [ NOT FOUND ]
      /etc/apache2/mods-enabled/negotiation.conf       [ NOT FOUND ]
      /etc/apache2/mods-enabled/setenvif.conf          [ NOT FOUND ]
      /etc/apache2/mods-enabled/mime.conf              [ NOT FOUND ]
      /etc/apache2/mods-enabled/reqtimeout.conf        [ NOT FOUND ]
      /etc/apache2/mods-enabled/dir.conf               [ NOT FOUND ]
      /etc/apache2/mods-enabled/evasive.conf           [ NOT FOUND ]
      /etc/apache2/mods-enabled/security2.conf         [ NOT FOUND ]
      /etc/apache2/mods-enabled/autoindex.conf         [ NOT FOUND ]
      /etc/apache2/mods-available/alias.conf           [ NOT FOUND ]
      /etc/apache2/mods-available/mpm_event.conf       [ NOT FOUND ]
      /etc/apache2/mods-available/proxy_ftp.conf       [ NOT FOUND ]
      /etc/apache2/mods-available/mpm_worker.conf      [ NOT FOUND ]
      /etc/apache2/mods-available/dav_fs.conf          [ NOT FOUND ]
      /etc/apache2/mods-available/proxy_balancer.conf  [ NOT FOUND ]
```

```
    /etc/apache2/mods-available/deflate.conf              [ NOT FOUND ]
    /etc/apache2/mods-available/ssl.conf                  [ NOT FOUND ]
    /etc/apache2/mods-available/ldap.conf                 [ NOT FOUND ]
    /etc/apache2/mods-available/status.conf               [ NOT FOUND ]
    /etc/apache2/mods-available/negotiation.conf          [ NOT FOUND ]
    /etc/apache2/mods-available/actions.conf              [ NOT FOUND ]
    /etc/apache2/mods-available/setenvif.conf             [ NOT FOUND ]
    /etc/apache2/mods-available/mime_magic.conf           [ NOT FOUND ]
    /etc/apache2/mods-available/http2.conf                [ NOT FOUND ]
    /etc/apache2/mods-available/cgid.conf                 [ NOT FOUND ]
    /etc/apache2/mods-available/cache_disk.conf           [ NOT FOUND ]
    /etc/apache2/mods-available/proxy_html.conf           [ NOT FOUND ]
    /etc/apache2/mods-available/mime.conf                 [ NOT FOUND ]
    /etc/apache2/mods-available/proxy.conf                [ NOT FOUND ]
    /etc/apache2/mods-available/reqtimeout.conf           [ NOT FOUND ]
    /etc/apache2/mods-available/dir.conf                  [ NOT FOUND ]
    /etc/apache2/mods-available/userdir.conf              [ NOT FOUND ]
    /etc/apache2/mods-available/evasive.conf              [ NOT FOUND ]
    /etc/apache2/mods-available/security2.conf            [ NOT FOUND ]
    /etc/apache2/mods-available/autoindex.conf            [ NOT FOUND ]
    /etc/apache2/mods-available/info.conf                 [ NOT FOUND ]
    /etc/apache2/mods-available/mpm_prefork.conf          [ NOT FOUND ]
    /etc/apache2/conf-enabled/serve-cgi-bin.conf          [ NOT FOUND ]
    /etc/apache2/conf-enabled/localized-error-pages.conf  [ NOT FOUND ]
    /etc/apache2/conf-enabled/charset.conf                [ NOT FOUND ]
    /etc/apache2/conf-enabled/security.conf               [ FOUND ]
    /etc/apache2/conf-enabled/other-vhosts-access-log.conf  [ NOT FOUND ]
    /etc/apache2/conf-enabled/javascript-common.conf      [ NOT FOUND ]
    /etc/apache2/sites-enabled/000-default.conf           [ NOT FOUND ]
    /etc/apache2/apache2.conf                             [ NOT FOUND ]
    /etc/apache2/sites-available/default-ssl.conf         [ NOT FOUND ]
    /etc/apache2/sites-available/000-default.conf         [ NOT FOUND ]
    /etc/apache2/conf-available/serve-cgi-bin.conf        [ NOT FOUND ]
    /etc/apache2/conf-available/localized-error-pages.conf  [ NOT FOUND ]
    /etc/apache2/conf-available/charset.conf              [ NOT FOUND ]
    /etc/apache2/conf-available/security.conf             [ FOUND ]
    /etc/apache2/conf-available/other-vhosts-access-log.conf  [ NOT FOUND ]
    /etc/apache2/conf-available/javascript-common.conf    [ NOT FOUND ]
    /etc/apache2/ports.conf                               [ NOT FOUND ]
  - Checking nginx                                        [ NOT FOUND ]

[+] SSH Support
------------------------------------
  - Checking running SSH daemon                           [ FOUND ]

[+] SNMP Support
------------------------------------
  - Checking running SNMP daemon                          [ NOT FOUND ]

[+] Databases
------------------------------------
  - PostgreSQL processes status                           [ FOUND ]
```

```
[+] LDAP Services
------------------------------------
  - Checking OpenLDAP instance                    [ NOT FOUND ]

[+] PHP
------------------------------------
  - Checking PHP                                  [ NOT FOUND ]

[+] Squid Support
------------------------------------
  - Checking running Squid daemon                 [ NOT FOUND ]

[+] Logging and files
------------------------------------
  - Checking for a running log daemon             [ OK ]
    - Checking Syslog-NG status                   [ NOT FOUND ]
    - Checking systemd journal status             [ FOUND ]
    - Checking Metalog status                     [ NOT FOUND ]
    - Checking RSyslog status                     [ FOUND ]
    - Checking RFC 3195 daemon status             [ NOT FOUND ]
    - Checking minilogd instances                 [ NOT FOUND ]
    - Checking wazuh-agent daemon status          [ NOT FOUND ]
  - Checking logrotate presence                   [ OK ]
  - Checking remote logging                       [ ENABLED ]
  - Checking log directories (static list)        [ DONE ]
  - Checking open log files                       [ DONE ]
  - Checking deleted files in use                 [ FILES FOUND ]

[+] Insecure services
------------------------------------
  - Installed inetd package                       [ NOT FOUND ]
  - Installed xinetd package                      [ OK ]
    - xinetd status                               [ NOT ACTIVE ]
  - Installed rsh client package                  [ OK ]
  - Installed rsh server package                  [ OK ]
  - Installed telnet client package               [ OK ]
  - Installed telnet server package               [ NOT FOUND ]
  - Checking NIS client installation              [ OK ]
  - Checking NIS server installation              [ OK ]
  - Checking TFTP client installation             [ OK ]
  - Checking TFTP server installation             [ OK ]

[+] Banners and identification
------------------------------------
  - /etc/issue                                    [ FOUND ]
    - /etc/issue contents                         [ OK ]
  - /etc/issue.net                                [ FOUND ]
    - /etc/issue.net contents                     [ OK ]

[+] Scheduled tasks
------------------------------------
  - Checking crontab and cronjob files            [ DONE ]
```

```
[+] Accounting
------------------------------------
  - Checking accounting information                    [ OK ]
  - Checking sysstat accounting data                   [ ENABLED ]
  - Checking auditd                                    [ ENABLED ]
    - Checking audit rules                             [ OK ]
    - Checking audit configuration file                [ OK ]
    - Checking auditd log file                         [ FOUND ]

[+] Time and Synchronization
------------------------------------
  - NTP daemon found: systemd (timesyncd)              [ FOUND ]
  - Checking for a running NTP daemon or client        [ OK ]
  - Last time synchronization                          [ 833s ]

[+] Cryptography
------------------------------------
  - Checking for expired SSL certificates [0/151]      [ NONE ]

  [WARNING]: Test CRYP-7902 had a long execution: 18.413721 seconds

  - Kernel entropy is sufficient                       [ YES ]
  - HW RNG & rngd                                      [ NO ]
  - SW prng                                            [ NO ]
  - MOR variable not found                             [ WEAK ]

[+] Virtualization
------------------------------------

[+] Containers
------------------------------------

[+] Security frameworks
------------------------------------
  - Checking presence AppArmor                         [ FOUND ]
    - Checking AppArmor status                         [ ENABLED ]
        Found 136 unconfined processes
  - Checking presence SELinux                          [ NOT FOUND ]
  - Checking presence TOMOYO Linux                     [ NOT FOUND ]
  - Checking presence grsecurity                       [ NOT FOUND ]
  - Checking for implemented MAC framework             [ OK ]

[+] Software: file integrity
------------------------------------
  - Checking file integrity tools
    - AIDE                                             [ FOUND ]
      - AIDE config file                               [ FOUND ]
      - AIDE database                                  [ FOUND ]
      - AIDE config (Checksum)                         [ OK ]
  - Checking presence integrity tool                   [ FOUND ]

[+] Software: System tooling
------------------------------------
```

```
  - Checking automation tooling
    - Ansible artifact                                  [ FOUND ]
  - Automation tooling                                  [ FOUND ]
  - Checking presence of Fail2ban                       [ FOUND ]
2026-04-19 19:52:24,785 fail2ban.configreader   [212692]: WARNING 'allowipv6' not defined in
'Definition'. Using default one: 'auto'
    - Checking Fail2ban jails                           [ ENABLED ]
  - Checking for IDS/IPS tooling                        [ FOUND ]

[+] Software: Malware
------------------------------------
  - Checking Rootkit Hunter                             [ FOUND ]
  - Malware software components                         [ FOUND ]
    - Active agent                                      [ NOT FOUND ]
    - Rootkit scanner                                   [ FOUND ]

[+] File Permissions
------------------------------------
  - Starting file permissions check
    File: /boot/grub/grub.cfg                           [ OK ]
    File: /etc/crontab                                  [ OK ]
    File: /etc/group                                    [ OK ]
    File: /etc/group-                                   [ OK ]
    File: /etc/hosts.allow                              [ OK ]
    File: /etc/hosts.deny                               [ OK ]
    File: /etc/issue                                    [ OK ]
    File: /etc/issue.net                                [ OK ]
    File: /etc/passwd                                   [ OK ]
    File: /etc/passwd-                                  [ OK ]
    File: /etc/ssh/sshd_config                          [ OK ]
    Directory: /root/.ssh                               [ OK ]
    Directory: /etc/cron.d                              [ OK ]
    Directory: /etc/cron.daily                          [ OK ]
    Directory: /etc/cron.hourly                         [ OK ]
    Directory: /etc/cron.weekly                         [ OK ]
    Directory: /etc/cron.monthly                        [ OK ]

[+] Home directories
------------------------------------
  - Permissions of home directories                     [ OK ]
  - Ownership of home directories                       [ OK ]
  - Checking shell history files                        [ OK ]

[+] Kernel Hardening
------------------------------------
  - Comparing sysctl key pairs with scan profile
    - dev.tty.ldisc_autoload (exp: 0)                   [ OK ]
    - fs.protected_fifos (exp: 2)                       [ OK ]
    - fs.protected_hardlinks (exp: 1)                   [ OK ]
    - fs.protected_regular (exp: 2)                     [ OK ]
    - fs.protected_symlinks (exp: 1)                    [ OK ]
    - fs.suid_dumpable (exp: 0)                         [ OK ]
    - kernel.core_uses_pid (exp: 1)                     [ OK ]
```

```
  - kernel.ctrl-alt-del (exp: 0)                          [ OK ]
  - kernel.dmesg_restrict (exp: 1)                        [ OK ]
  - kernel.kptr_restrict (exp: 2)                         [ OK ]
  - kernel.modules_disabled (exp: 1)                      [ OK ]
  - kernel.perf_event_paranoid (exp: 2 3 4)               [ OK ]
  - kernel.randomize_va_space (exp: 2)                    [ OK ]
  - kernel.sysrq (exp: 0)                                 [ OK ]
  - kernel.unprivileged_bpf_disabled (exp: 1)             [ OK ]
  - kernel.yama.ptrace_scope (exp: 1 2 3)                 [ OK ]
  - net.core.bpf_jit_harden (exp: 2)                      [ OK ]
  - net.ipv4.conf.all.accept_redirects (exp: 0)           [ OK ]
  - net.ipv4.conf.all.accept_source_route (exp: 0)        [ OK ]
  - net.ipv4.conf.all.bootp_relay (exp: 0)                [ OK ]
  - net.ipv4.conf.all.forwarding (exp: 0)                 [ OK ]
  - net.ipv4.conf.all.log_martians (exp: 1)               [ OK ]
  - net.ipv4.conf.all.mc_forwarding (exp: 0)              [ OK ]
  - net.ipv4.conf.all.proxy_arp (exp: 0)                  [ OK ]
  - net.ipv4.conf.all.rp_filter (exp: 1)                  [ OK ]
  - net.ipv4.conf.all.send_redirects (exp: 0)             [ OK ]
  - net.ipv4.conf.default.accept_redirects (exp: 0)       [ OK ]
  - net.ipv4.conf.default.accept_source_route (exp: 0)    [ OK ]
  - net.ipv4.conf.default.log_martians (exp: 1)           [ OK ]
  - net.ipv4.icmp_echo_ignore_broadcasts (exp: 1)         [ OK ]
  - net.ipv4.icmp_ignore_bogus_error_responses (exp: 1)   [ OK ]
  - net.ipv4.tcp_syncookies (exp: 1)                      [ OK ]
  - net.ipv4.tcp_timestamps (exp: 0 1)                    [ OK ]
  - net.ipv6.conf.all.accept_redirects (exp: 0)           [ OK ]
  - net.ipv6.conf.all.accept_source_route (exp: 0)        [ OK ]
  - net.ipv6.conf.default.accept_redirects (exp: 0)       [ OK ]
  - net.ipv6.conf.default.accept_source_route (exp: 0)    [ OK ]

[+] Hardening
------------------------------------
  - Installed compiler(s)                                 [ FOUND ]
  - Installed malware scanner                             [ FOUND ]
  - Non-native binary formats                             [ FOUND ]

[+] Custom tests
------------------------------------
  - Running custom tests...                               [ NONE ]

[+] Plugins (phase 2)
------------------------------------

================================================================================
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  -[ Lynis 3.1.6 Results ]-

  Great, no warnings

  Suggestions (3):
  ----------------------------
```

```
  * Consider hardening system services [BOOT-5264]
    - Details  : Run '/usr/bin/systemd-analyze security SERVICE' for each service
    - Related resources
      * Article: Systemd features to secure service files: https://linux-audit.com/systemd/systemd-
features-to-secure-units-and-services/
      * Website: https://cisofy.com/lynis/controls/BOOT-5264/

  * Check iptables rules to see which rules are currently not used [FIRE-4513]
    - Related resources
      * Website: https://cisofy.com/lynis/controls/FIRE-4513/

  * Check what deleted files are still in use and why. [LOGG-2190]
    - Related resources
      * Website: https://cisofy.com/lynis/controls/LOGG-2190/

  Follow-up:
  ----------------------------
  - Show details of a test (lynis show details TEST-ID)
  - Check the logfile for all details (less /var/log/lynis.log)
  - Read security controls texts (https://cisofy.com)
  - Use --upload to upload data to central system (Lynis Enterprise users)

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  Lynis security scan details:

  Scan mode:
  Normal [■]  Forensics [ ]  Integration [ ]  Pentest [ ]

  Lynis modules:
  - Compliance status      [?]
  - Security audit         [V]
  - Vulnerability scan     [V]

  Details:
  Hardening index : 90 [##################  ]
  Tests performed : 262
  Plugins enabled : 0

  Software components:
  - Firewall               [V]
  - Intrusion software     [V]
  - Malware scanner        [V]

  Files:
  - Test and debug information      : /var/log/lynis.log
  - Report data                     : /var/log/lynis-report.dat

================================================================================
=======

  Lynis 3.1.6
```

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2025, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

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[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)